

OpenSCAD SOFTWARE For 3D Modelling

Designing prototypes is made easier by creating the model digitally first. This ensures accuracy and flawlessness in the design before moving ahead with the physical creation. For this purpose, computer-aided design (CAD) software becomes highly beneficial. OpenSCAD is a great option in this category. It supports 3D solid-modelling, which provides

an extra sense of accuracy towards the design.

OpenSCAD, an overview

OpenSCAD is a free open source software, licensed under GPL v2, compatible across multiple major platforms including Windows, Mac OS and Linux/Unix.

Developed in C++, its operations are completely script-based. Hence,

you will have to type in the appropriate command script. OpenSCAD comes with an in-built script editor and compiler for the purpose. The scripting language is simple and mostly follows equation models. Therefore the whole process comes down to mathematical flows, leaving a small learning curve for the scripting language. The language supports variables and parameters, basic shapes for design creation like cube, cylinder and sphere, and a large number of functions like intersection, union, translate and hull.

The user interface is very simple, and is developed on Qt. The main work area is a two-screen window, one for editing and compiling scripts, and the other for the main visualisation. On the top, there is a menu bar that includes five menus, namely, File, Edit, Design, View and Help. Different operations can be initiated through the options under these menus.

The left side of the screen is used for writing and compiling the syntax.

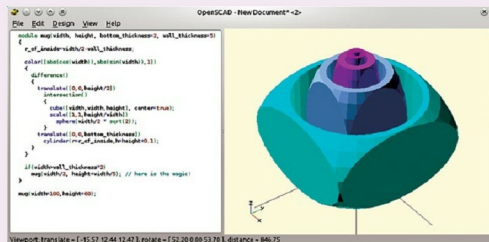


Fig. 1: OpenSCAD user interface (Credit: openscad.org)

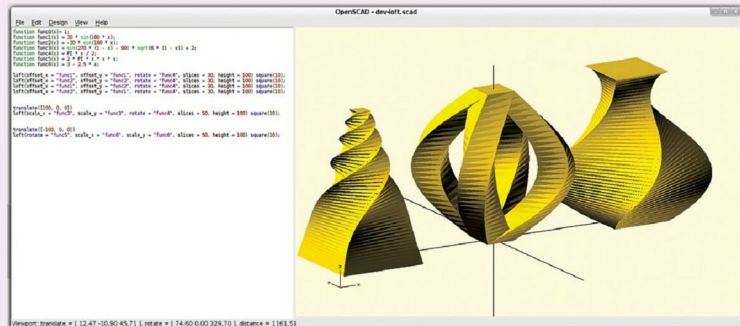


Fig. 2: Compiling a design (Credit: github.com)